

ALGORITHM

1. Start
2. Define decorator function **calculator(choice)**
 - Accepts the user's operation choice as input.
3. Inside **calculator(choice)**, define another function **calculator(func)**
 - This function will receive the original function (calculate).
4. Define inner function **inner()**
 - Call the original function to get two numbers a and b.
5. Check the user's choice
 - If **choice is 1**, return $a + b$ (Addition).
 - If **choice is 2**, return $a - b$ (Subtraction).
 - If **choice is 3**:
 - If b is 0, return "**Cannot divide by zero**".
 - Else, return a / b (Division).
 - Otherwise, return "**Invalid choice**".
6. Return the inner function
 - This replaces the original function behavior using a decorator.
7. Prompt the user to enter two numbers
 - Read both numbers in the same line and store them as x and y.
8. Display menu for operation choice
 - Show options for Addition, Subtraction, and Division.
9. Read the user's choice

- Store it in the variable choice.

10. Apply the decorator **@calculator(choice)** to the function calculate

- The function returns the two input numbers x and y.

11. Call the decorated function **calculate()**

- The selected arithmetic operation is performed.

12. Display the result

- Print the output in the format: **Result:_____**.

13. End